SpaceX Falcon 9
First-stage landing prediction

Applied Data Science Capstone
Course dataset analysis

Executive summary

Used Python, SQLite, charts, Folium and Dash to explore launch and landing records. Compared four classifiers with a majority baseline.

60 of 90 first stages were labeled successful (66.7%). CCAFS SLC 40 hosted 55 launches.

The decision tree led training cross-validation (75%) and achieved 77.8% on the 18-record holdout.

The sample is small and historical; associations do not establish cause.

Source: IBM Skills Network course dataset

Introduction

Falcon 9 is designed to recover and reuse its first stage.

Question: Can mission information help predict whether the first stage lands successfully?

Analysis covers launch sites, payloads, orbit, time and reuse, followed by classification.

The target describes landing outcome, not whether the payload mission succeeded.

Source: IBM Skills Network course dataset

Dataset and scope

90 Falcon 9 records from 4 June 2010 through 5 November 2020.

Class 1: successful first-stage landing or controlled ocean outcome.
Class 0: unsuccessful or no landing, including no attempt.

Course-prepared payload data includes imputed values.

A separate 101-row course CSV supports the SQL section. Counts across the two files are not directly comparable.

Source: IBM Skills Network course dataset

Data collection

The 90-record analysis uses an IBM course-prepared Falcon 9 CSV.

A separate 101-row course CSV supports the SQL exercises.

The following slides demonstrate API-response parsing and Wikipedia scraping. These samples are not joined to the analysis datasets.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/data_collection.py

Data collection: SpaceX API

1. Request past-launch records from the SpaceX v4 API.

2. Decode JSON and extract flight number, date, rocket and outcome fields.

3. Save tabular output for inspection.

The live API request timed out during this run, so the script used the IBM course snapshot: 107 records. This sample is separate from the 90-row modeling file.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/data_collection.py

Data collection: Wikipedia scraping

1. Request the fixed 9 June 2021 Wikipedia revision.

2. Parse launch tables using BeautifulSoup and extract complete ten-column rows.

3. Save the extracted values to a CSV file.

Result: 9 tables and 121 complete launch rows. The scrape is a separate demonstration, not the source of the model results.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/data_collection.py

Data wrangling

Kept launch site, date, orbit, payload mass and flight-related features.

Checked class counts and launch-site distribution.

For modeling: split before training, then impute, scale numeric variables and one-hot encode categories inside each training fold.

The supplied dataset has already filled some missing payload values.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/spacex_capstone.ipynb

Exploratory analysis methods

Scatter plots compare payload mass with landing outcome.

Bar charts compare launch counts and success rates across sites and orbits.

A yearly chart shows how the observed success rate changes over time.

Group sizes accompany rates; comparisons show association, not causation.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/spacex_capstone.ipynb

SQL analysis methods

Loaded the separate SpaceX CSV into a local SQLite table.

Used SELECT, WHERE, GROUP BY and aggregation to inspect sites, payloads and outcomes.

Preserved source site labels, including CCAFS LC-40 and CCAFS SLC-40.

The SQL CSV has 101 records, versus 90 in the modeling file.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/sql_extra.py

Interactive visual analytics methods

Folium: plot launch-site locations and 90 clustered launch records, with an example distance to a coastal reference point.

Plotly Dash: filter by launch site and payload mass, then view landing outcomes and the payload scatter plot.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/visual_analytics.py

Predictive analysis methods

Target: first-stage landing class.

Features: payload, flights, reuse count, block, reused status, orbit and launch site.

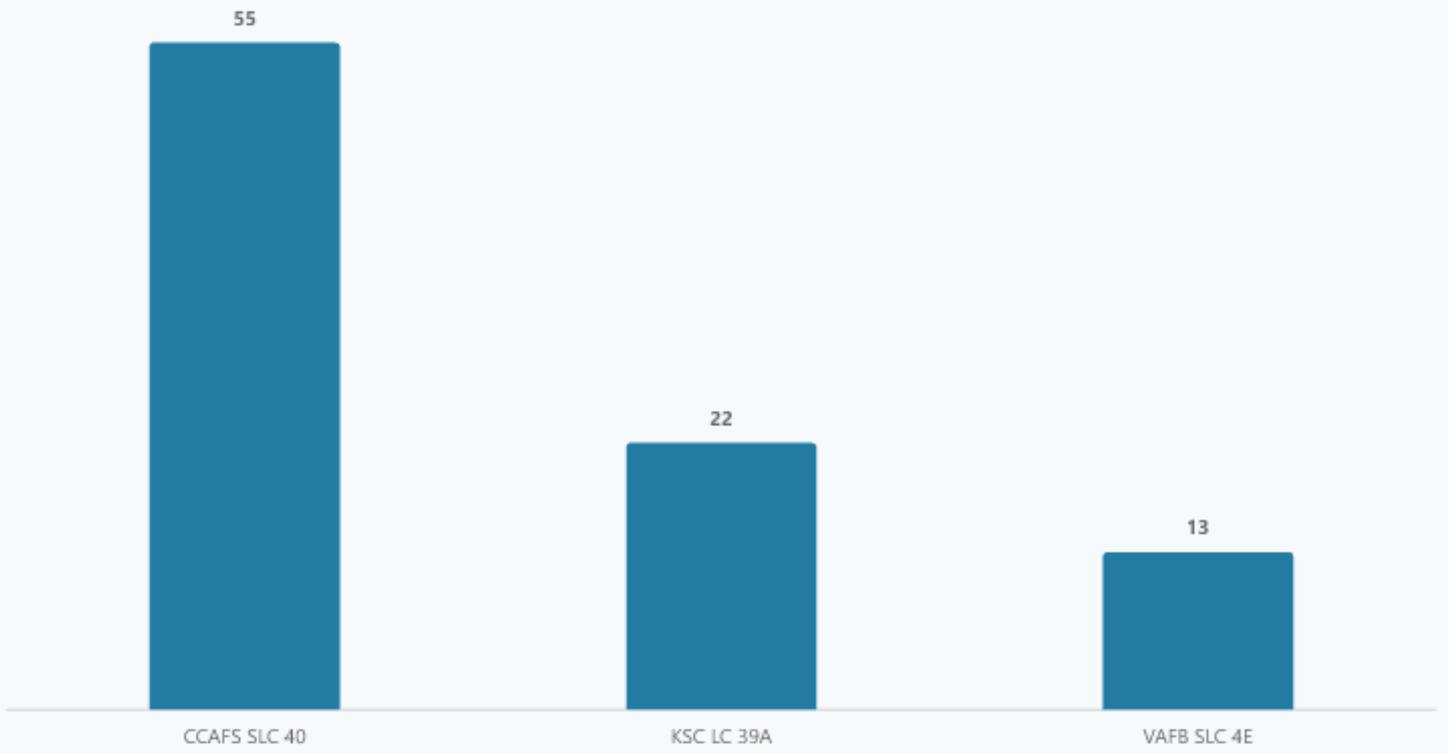
Stratified 80/20 split: 72 training records and 18 test records.

Compared a majority baseline with logistic regression, SVM, decision tree and KNN using four-fold training cross-validation.

No hyperparameter search was performed.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/analyze.py

CCAFS SLC 40 hosted 55 of 90 launches

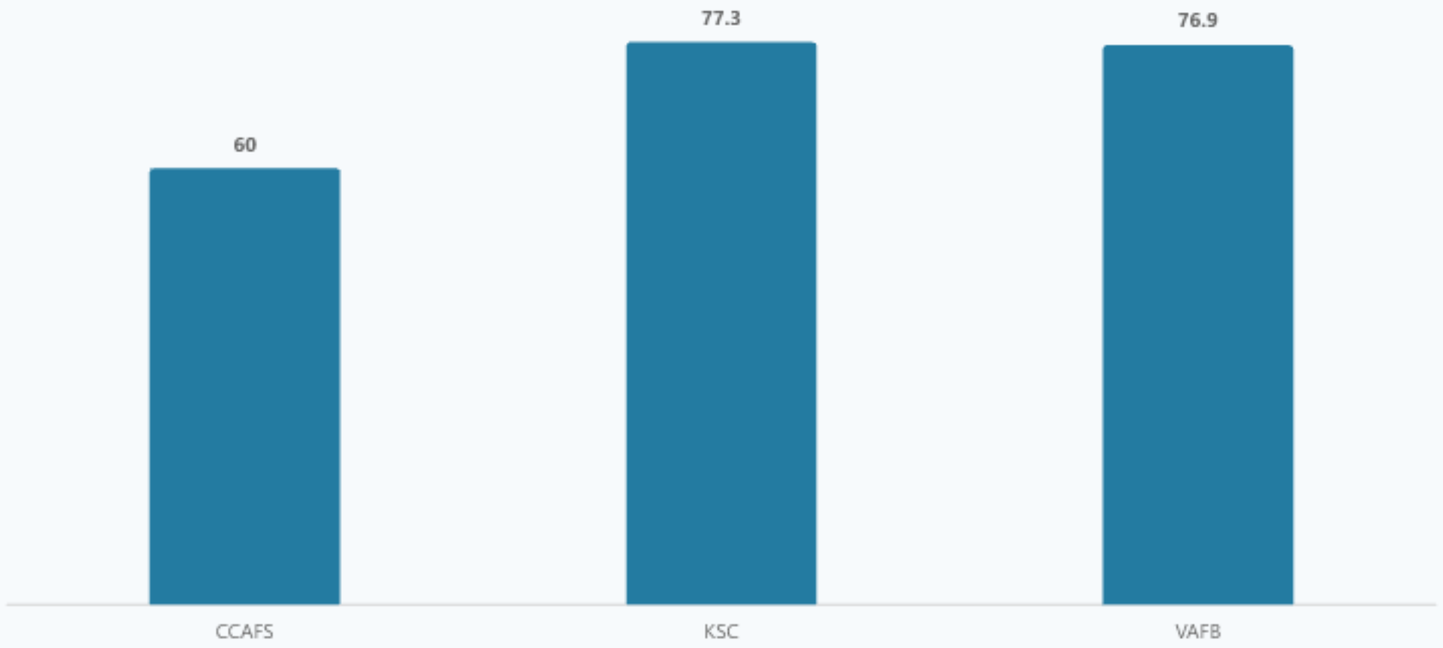


Launch counts differ substantially by site.

The CCAFS sample includes 33 successful landings out of 55 launches.

Source: IBM Skills Network course dataset (90 records)

Landing success differed across launch sites

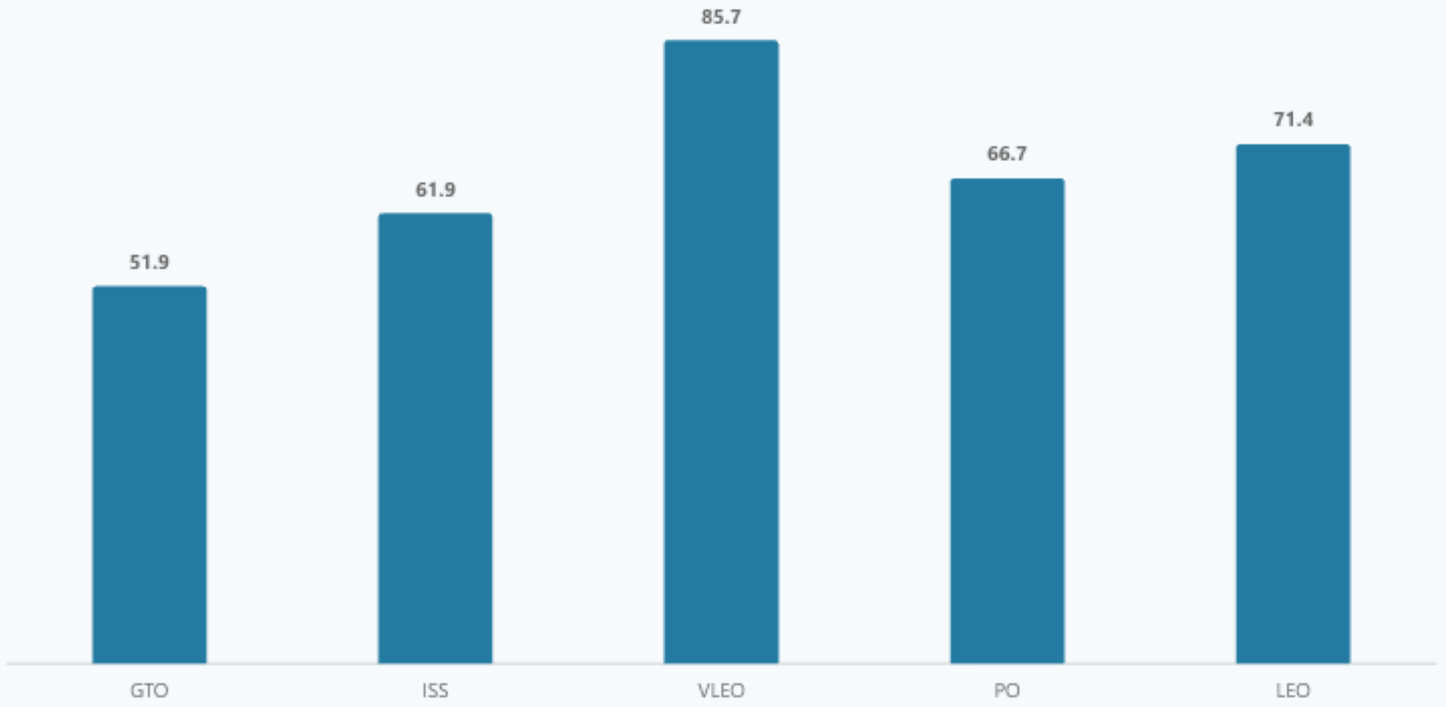


CCAFS: 33 / 55
KSC: 17 / 22
VAFB: 10 / 13

These historical differences do not establish that site caused the outcome.

Source: IBM Skills Network course dataset (90 records)

Success rates varied by orbit

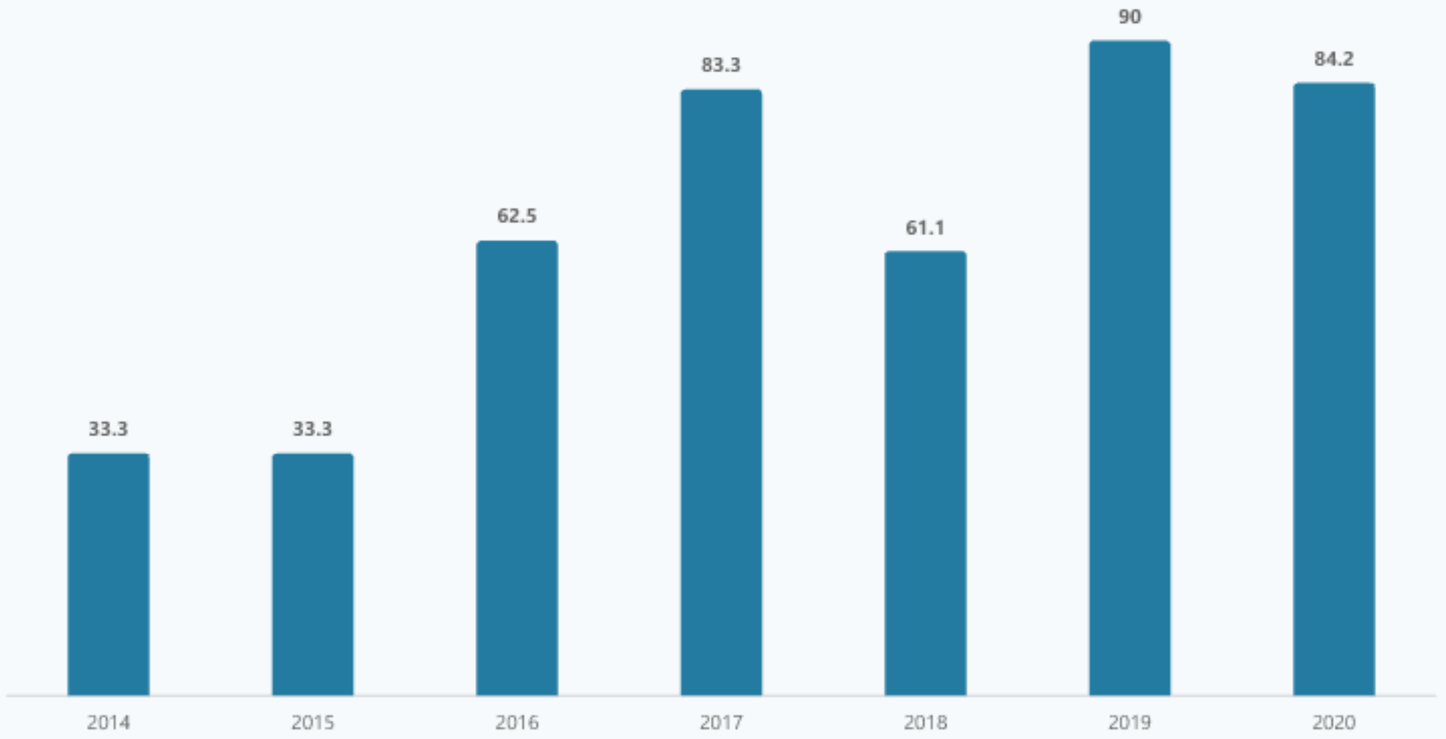


GTO: 14 / 27
ISS: 13 / 21
VLEO: 12 / 14

Orbit groups differ in size and launch conditions.

Source: IBM Skills Network course dataset (90 records)

Later years generally had higher success rates

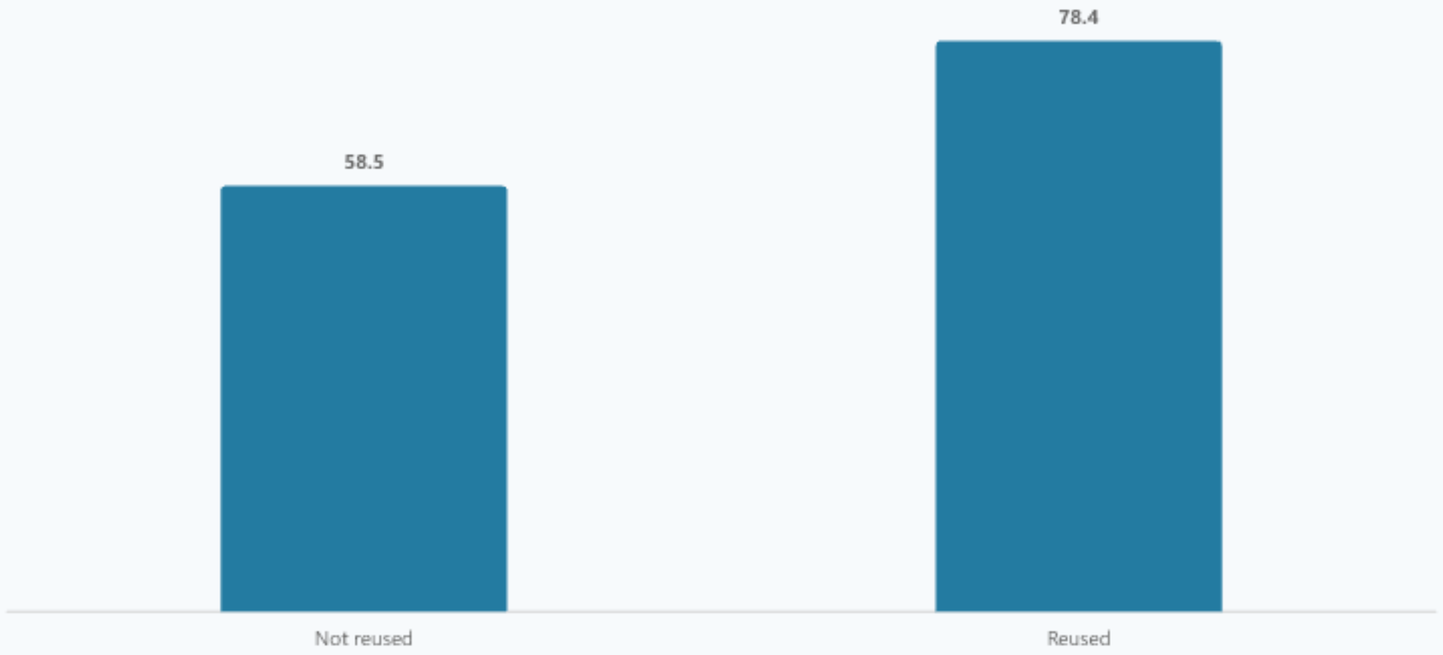


2014: 2 / 6
2017: 15 / 18
2020: 16 / 19

Rates reflect a selected dataset and small yearly samples.

Source: IBM Skills Network course dataset (90 records)

Reused boosters had a higher observed landing rate



Not reused: 31 / 53
Reused: 29 / 37

Booster design and launch year may also differ across groups. The comparison is not causal.

Source: IBM Skills Network course dataset (90 records)

SQL results: launch sites

```
SELECT Launch_Site, COUNT(*) FROM launches
GROUP BY Launch_Site ORDER BY COUNT(*) DESC;
```

CCAFS SLC-40: 34 CCAFS LC-40: 26
KSC LC-39A: 25 VAFB SLC-4E: 16

Labels in the SQL source are retained as recorded.

Source: IBM SQL course CSV (101 records)

SQL results: NASA payload and Falcon 9 v1.1

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM launches
WHERE Customer = 'NASA (CRS)';
```

Total NASA (CRS) payload: 45,596 kg

F9 v1.1 average payload: 2,534.67 kg
Calculated from rows with booster version beginning “F9 v1.1”.

Source: IBM SQL course CSV (101 records)

SQL results: landing and payload

First recorded successful ground-pad landing: 22 Dec 2015.

Maximum payload mass in this SQL file: 15,600 kg. The maximum occurs on multiple rows.

Landing outcome categories include 14 explicitly labeled “Success (drone ship)”.

The separate 90-row modeling data uses a different outcome coding.

Source: IBM SQL course CSV (101 records)

SQL results: mission outcomes and boosters

After trimming whitespace, mission outcomes in the 101-row SQL file are:
99 exact “Success” (98.0%), 1 in-flight failure, and 1 success with unclear payload status.

Four drone-ship successes carried 4,000–6,000 kg:
B1022 (4,696), B1026 (4,600), B1021.2 (5,300), B1031.2 (5,200).

Mission success is distinct from first-stage landing success.

Source: IBM SQL course CSV (101 records)

SQL results: early landing outcomes

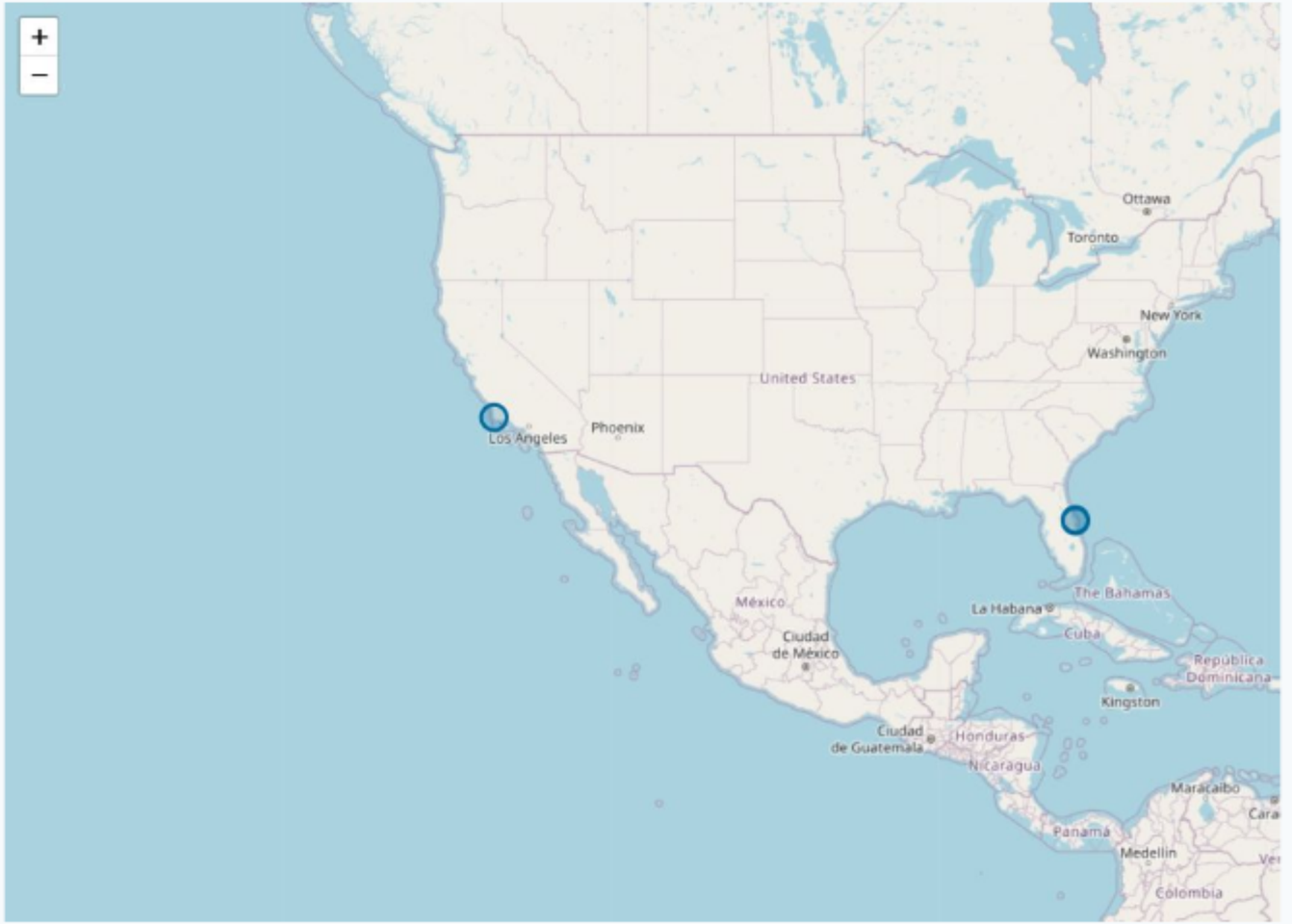
Two drone-ship landing failures in 2015:
10 Jan: B1012, CCAFS LC-40
14 Apr: B1015, CCAFS LC-40

From 4 Jun 2010 through 20 Mar 2017:
10 no attempt, 5 drone-ship successes and 5 drone-ship failures.

First five CCA-prefixed rows in the SQL file are CCAFS LC-40 launches.

Source: IBM SQL course CSV (101 records)

Folium map: launch sites



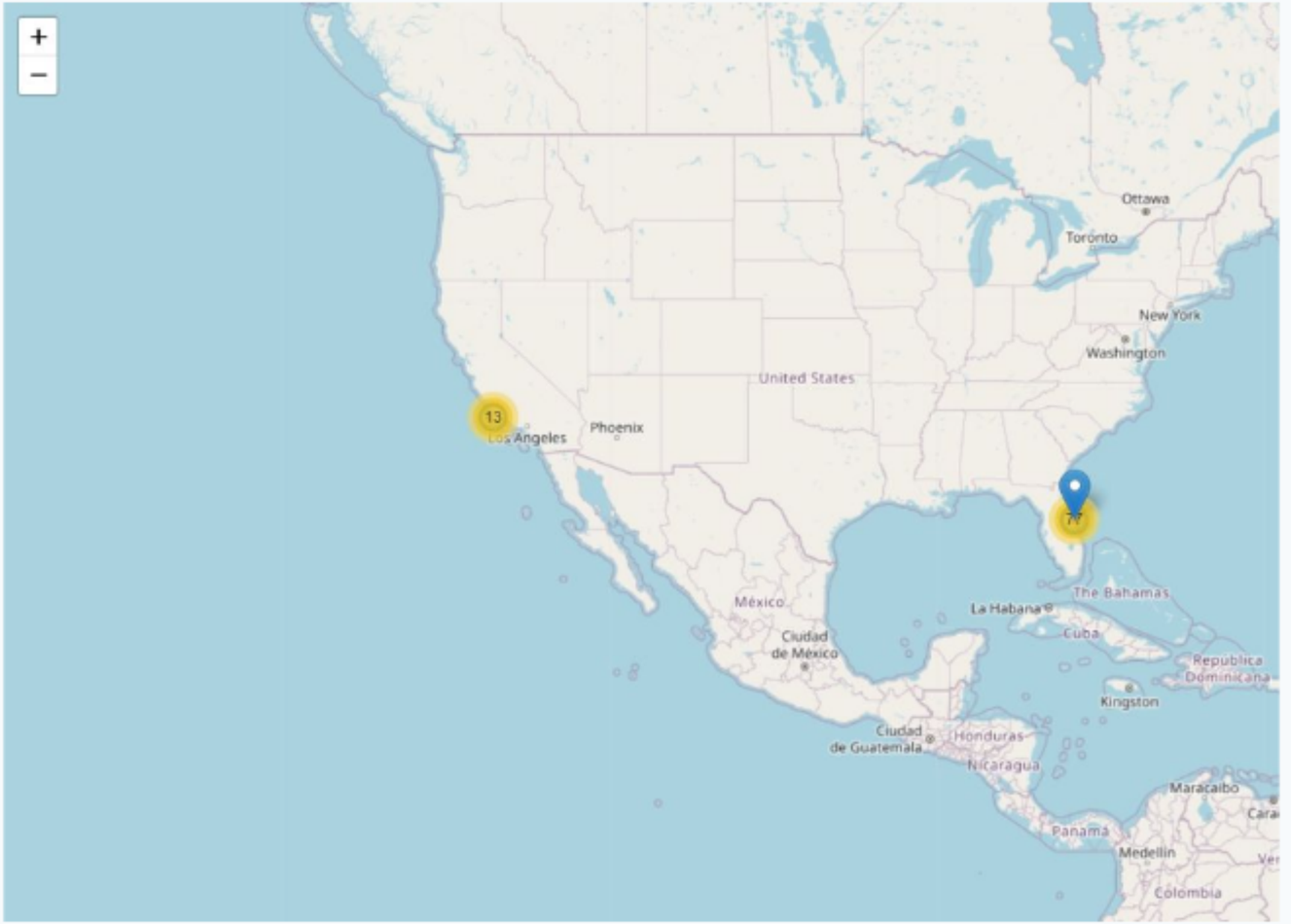
The dataset places two sites in Florida and one in California.

Markers show location; the two Florida sites overlap at this zoom.

Interactive map popups show launch counts and successful landings.

Data: IBM course dataset. Map © OpenStreetMap contributors.

Folium map: all launch records

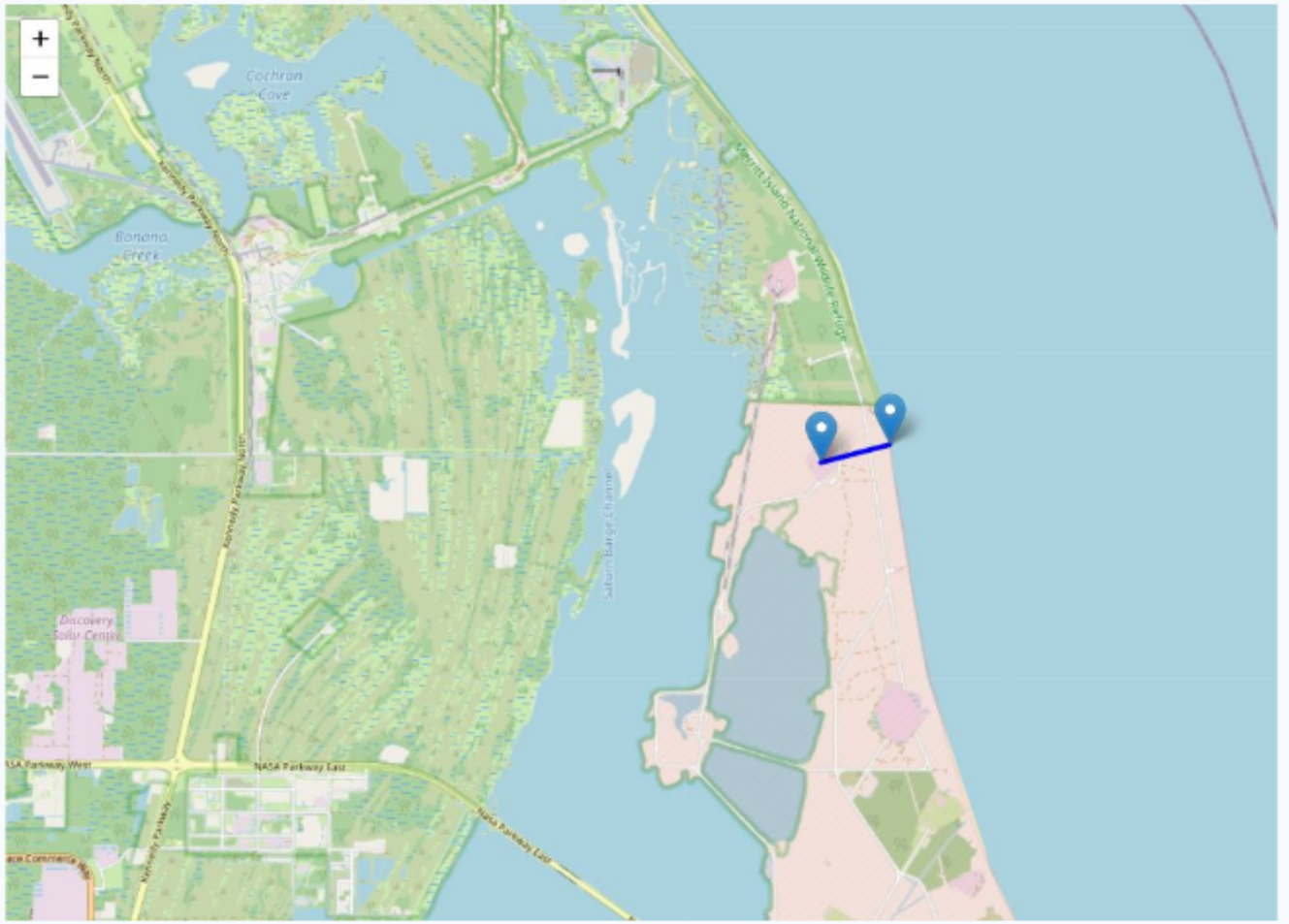


Each of the 90 records has a marker. Clusters show 77 launches in Florida and 13 in California.

Open a cluster and marker to inspect the flight date and landing class.

Map © OpenStreetMap contributors. Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/visual_analytics.py

Proximity to an approximate coastal point

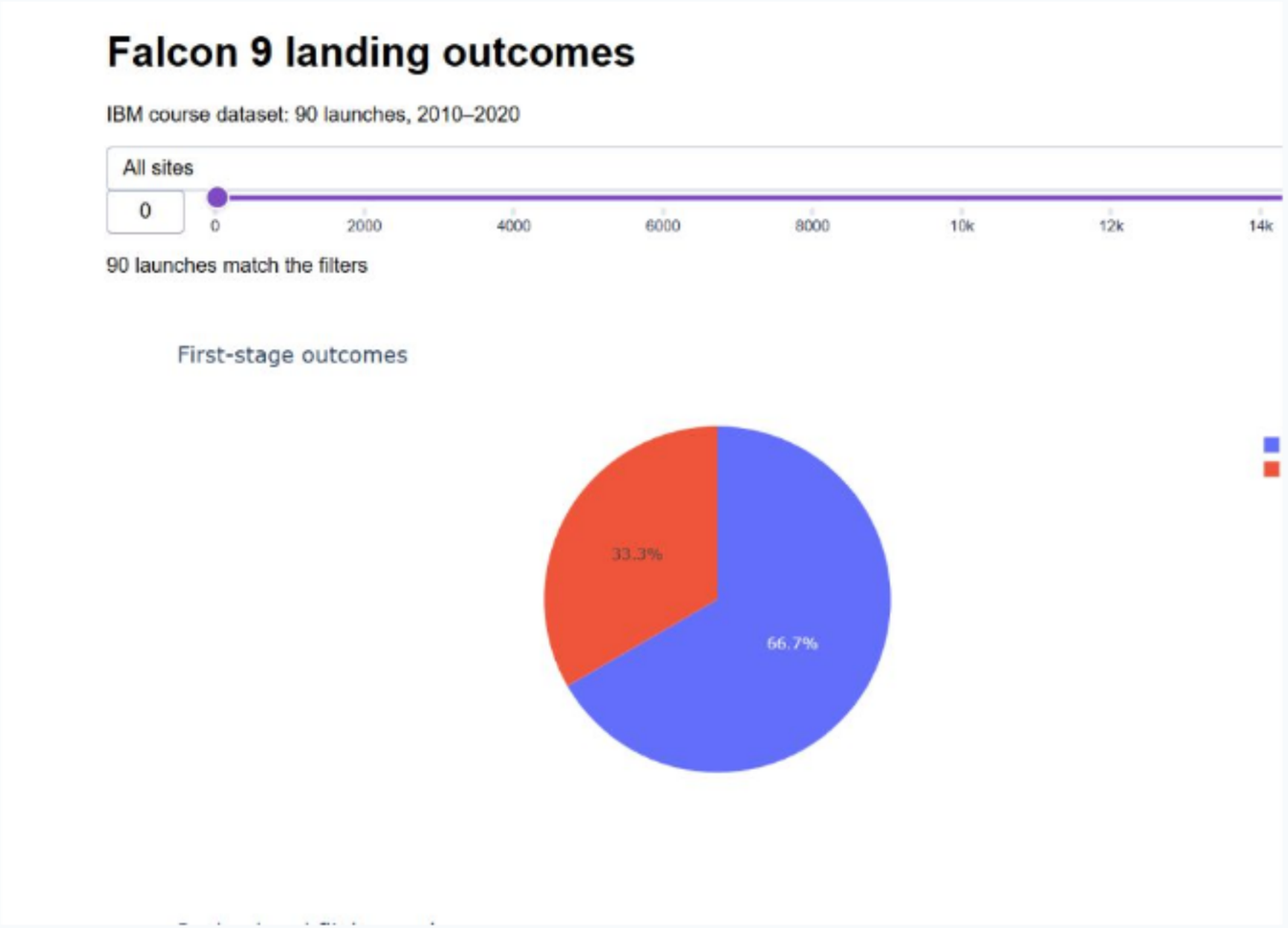


CCAFS SLC 40 to the selected coastal reference point: approximately 0.94 km in a straight line.

This reference point is approximate. It is not a measurement of the nearest shoreline.

Map © OpenStreetMap contributors. Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/visual_analytics.py

Dash dashboard: exploring outcomes



The default view includes all 90 records.

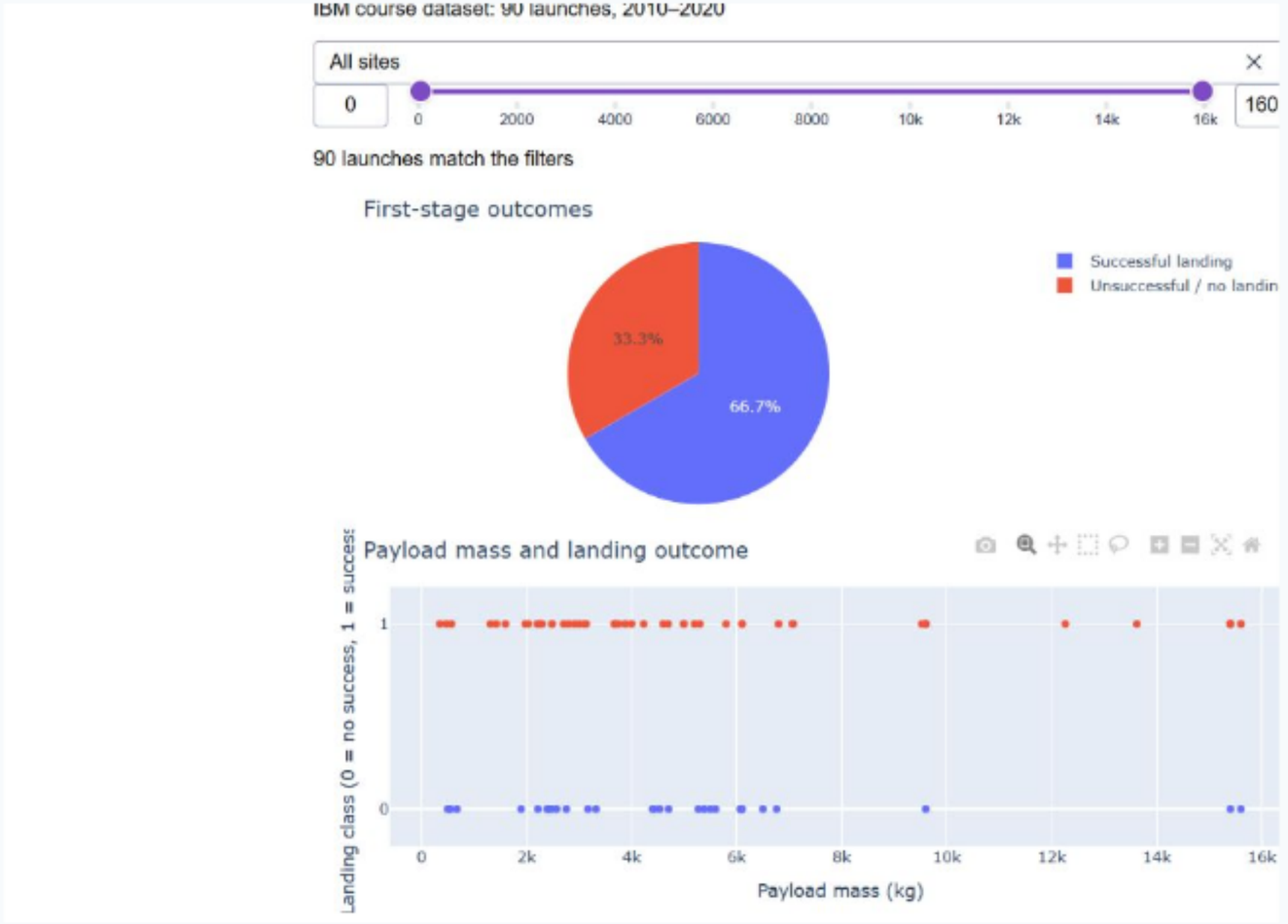
60 successful landings: 66.7%.

Site and payload filters update the outcome chart and scatterplot together.

Static screenshot of a locally run app.

Source: IBM Skills Network course dataset

Dash: payload mass and landing outcome

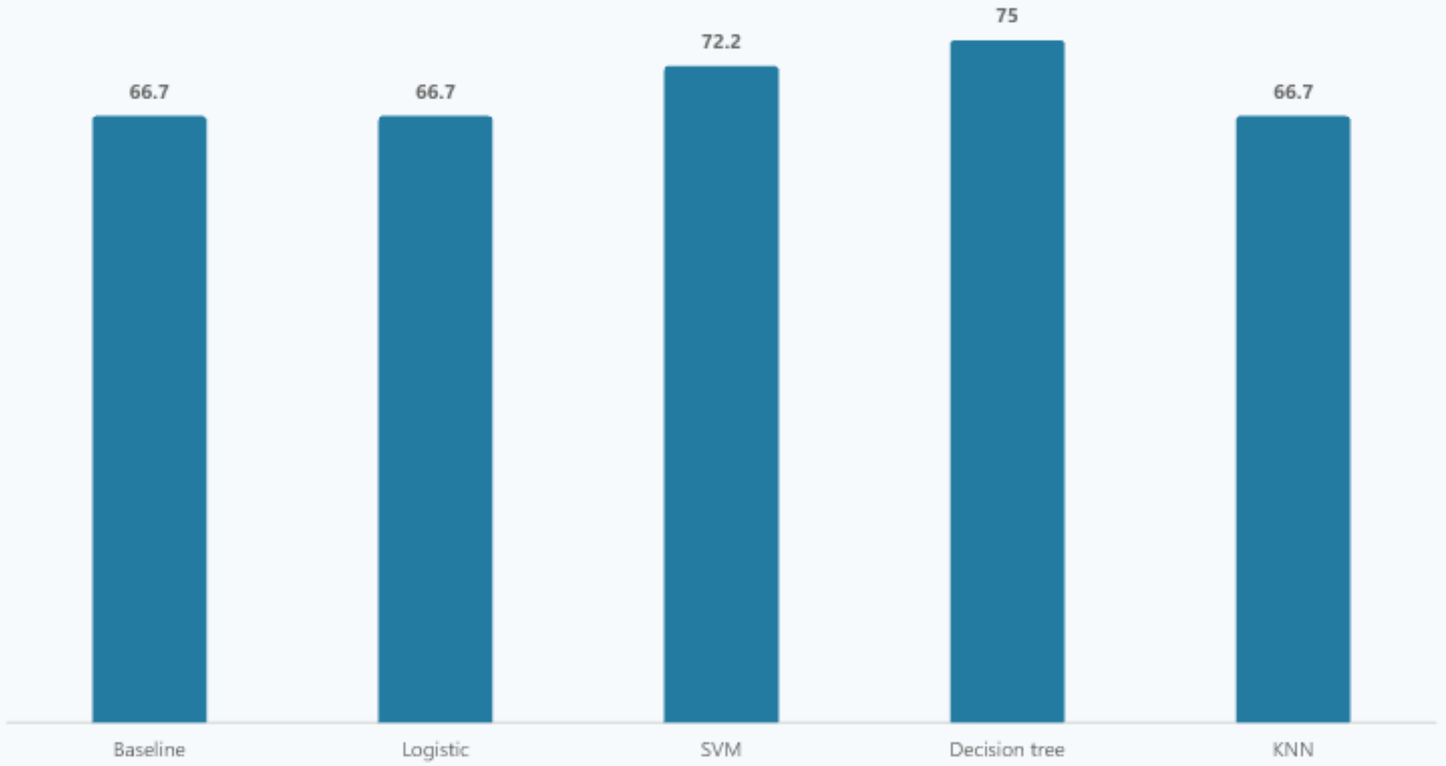


The all-site view shows the success pie chart and payload-versus-outcome scatter plot.

A dot is a launch record; repeated values overlap. Site and payload filters update both figures.

Code: https://github.com/sabakan0123/spacex-falcon9-capstone/blob/main/visual_analytics.py

Decision tree led training cross-validation



Four-fold cross-validation on 72 training records.

The decision tree had the highest mean score.

The differences are uncertain with this small sample.

Source: IBM Skills Network course dataset (90 records)

Decision tree: holdout results

Selected using training cross-validation, the decision tree correctly classified 14 of 18 holdout records (77.8%).

Confusion matrix	Predicted no	Predicted yes
Actual no landing	3	3
Actual successful landing	1	11

The model missed half of the six non-successes in this test split.

Source: local analysis, 18-record stratified holdout; seed 42

Model comparison and uncertainty

Holdout accuracy: logistic regression 88.9%, SVM 77.8%, decision tree 77.8%, KNN 77.8%, majority baseline 66.7%.

The tree was selected on training cross-validation, not on the holdout result.

A single test split of 18 records is unstable. Further time-based testing on newer launches is needed before operational use.

Source: local analysis; fixed settings, no hyperparameter search

Conclusion

Landing success increased in later years within this course sample.

Site, orbit and reused status show differences, but timing and mission characteristics may confound them.

The selected decision tree surpassed the majority baseline on this holdout, while missing 3 of 6 non-successes.

Next: test on later launches and verify that each predictor is available before launch.

Source: IBM Skills Network course dataset

Sources and reproducibility

IBM Skills Network: dataset_part_2.csv (90 launches)
cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_2.csv

IBM Skills Network: Spacex.csv (101 SQL records)
cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/labs/module_2/data/Spacex.csv

Code: analyze.py, sql_extra.py and visual_analytics.py

Repository: https://github.com/sabakan0123/spacex-falcon9-capstone

Folium map tiles © OpenStreetMap contributors